LiverPath AI

Uncertainty-Aware Deep Learning for Multiclass Liver Histopathology Patch Classification

Scientific research report | Exploratory computational pathology study

6 November 2026 • Frozen patch-level test evaluation

Research scope: seven-class H&E liver image patch classification, uncertainty-guided error prioritisation, and post-hoc probability calibration. Not a clinical diagnostic system.

Abstract

Deep learning can support histopathology image analysis, but classification accuracy alone does not establish whether a model's predictions are reliable. This exploratory study investigated seven-class liver histopathology patch classification using transfer learning, selective fine-tuning, predictive entropy and temperature scaling. HepatoBench provided 87,301 original 150 × 150 RGB H&E image entries across fibrosis, inflammation, necrosis, normal tissue, biliary reaction, steroidal degeneration and tumour. Following removal of one unreadable image and 80 exact duplicates, 87,220 unique patches were partitioned into stratified training, validation and test subsets. An ImageNet-pretrained ResNet-18 with a frozen backbone served as a baseline; the final residual block and classification head were subsequently fine-tuned. The final checkpoint and evaluation choices were frozen before held-out test inference. On 13,083 test patches, the fine-tuned model achieved 92.92% accuracy, 91.34% balanced accuracy and 0.9005 macro-F1. Uncalibrated predictive entropy identified 56.80% of 926 errors by prioritising the 10% most uncertain patches (error-detection AUROC 0.9079). Temperature scaling modestly reduced 15-bin expected calibration error from 0.0084 to 0.0062. Tumour–normal confusion and residual errors outside the review subset remained important limitations. Patient and slide identifiers were unavailable, so the results do not establish patient-independent generalisation or clinical suitability.

Keywords: computational pathology; liver histopathology; ResNet-18; transfer learning; predictive entropy; temperature scaling; PyTorch.

1. Introduction

1.1 Background and motivation

Liver histopathology encompasses diverse microscopic appearances, including fibrosis, inflammation, necrosis, degenerative changes and tumour-associated alterations. Convolutional neural networks can learn image representations that separate such categories, while pretrained networks can reduce the resources required for specialised image classification. Yet even models with high average accuracy may misclassify visually ambiguous patches with high confidence. Assessing uncertainty and calibration is therefore an important complement to conventional classification metrics.

1.2 Research questions

- Can ImageNet-based transfer learning and selective fine-tuning classify seven HepatoBench tissue categories effectively?
- Can predictive entropy prioritise misclassified patches under a fixed, limited review budget?
- Does validation-fitted temperature scaling improve confidence calibration on a held-out patch-level test split?

1.3 Scope

The study evaluates image patches, not whole slides or patient diagnoses. Uncertainty-guided "review" refers to retrospective error prioritisation against known labels; no pathologist review or correction experiment was performed.

2. Dataset and preparation

2.1 Dataset composition

The HepatoBench dataset was obtained from the Hugging Face repository xtxx/HepatoBench, pinned to revision 3fd9d3697fb63a84ea0635aed97f77ad3f1c5d97. The original archive contained 87,301 entries of 150 × 150 RGB H&E image patches. The dataset's STE label denotes steroidal degeneration, not steatosis.

Class	Category	Original entries
FIB	Fibrosis	26,364
INF	Inflammation	4,776
NEC	Necrosis	22,023
NOR	Normal tissue	13,200
REA	Biliary reaction	1,540
STE	Steroidal degeneration	6,198
TUM	Tumour	13,200
Total		87,301

2.2 Image integrity and duplicate control

One zero-byte fibrosis image (FIB/02_FIB_3355.png) was excluded, leaving 87,300 valid images. SHA-256 hashing identified 87,220 unique image hashes and 80 exact duplicate pairs, all within FIB. Only unique representatives were retained for model splitting, preventing cross-split exact-hash duplication. Perceptual hashes were also inspected; no candidate pairs within Hamming distance five were found among the SHA-unique images. These checks cannot exclude patches originating from related slide regions.

2.3 Data partitions

A fixed-seed (42), stratified, patch-level split allocated approximately 70% to training, 15% to validation and 15% to final testing. The validation split supported checkpoint selection, exploratory error analysis and calibration development. The test split was evaluated only after freezing the final protocol.

Class	Train	Validation	Test
FIB	18,398	3,942	3,943
INF	3,344	716	716
NEC	15,416	3,304	3,303
NOR	9,240	1,980	1,980
REA	1,078	231	231
STE	4,338	930	930
TUM	9,240	1,980	1,980
Total	61,054	13,083	13,083

2.4 Preprocessing

Images were resized to 224 × 224 and normalised using ImageNet channel statistics. Training transformations included random horizontal and vertical flips, rotations up to 15°, and colour jitter;

validation and test transformations were deterministic. Class imbalance motivated weighted cross-entropy and macro-averaged evaluation metrics.

3. Methodology

3.1 Experimental Framework

LiverPath AI was developed as an exploratory deep-learning framework for multiclass liver histopathology patch classification. The workflow comprised dataset quality control and stratified splitting; baseline transfer learning; selective fine-tuning; uncertainty estimation and error analysis; temperature scaling; and frozen held-out test evaluation.

The framework was implemented in Python with PyTorch and torchvision. Training and inference used Apple Silicon hardware through the Metal Performance Shaders (MPS) backend.

3.2 Model Architecture: ResNet-18

ResNet-18 was selected as a relatively lightweight convolutional neural network with residual skip connections and established transfer-learning support. The ImageNet-pretrained classification layer was replaced by a seven-output fully connected layer for fibrosis (FIB), inflammation (INF), necrosis (NEC), normal tissue (NOR), biliary reaction (REA), steroidal degeneration (STE) and tumour (TUM).

The model produces seven logits for each image. Softmax converts these logits into predicted class probabilities:

$$p(y=k \vee x)=\exp(z_k)/\sum_{j=1}^{7}\exp(z_j)$$

Here z_k is the logit for class k, and the predicted class is the one with the largest softmax probability.

3.3 Baseline Transfer-Learning Model

The baseline used an ImageNet-pretrained ResNet-18 with the convolutional backbone frozen and only the new classification layer trained. The model had approximately 11.18 million parameters, of which 3,591 were trainable. It was optimised for five epochs with AdamW (learning rate 0.001) and class-weighted cross-entropy to address unequal class frequencies.

The class-weighted cross-entropy principle can be expressed as:

$$L=-(1/N)\sum_{i=1}^{N} w_{yi}\log p(y_i \vee x_i)$$

N is the number of examples and w_{yi} is the weight for the true class of example i. The expression illustrates the weighting principle; the implementation used PyTorch weighted cross-entropy. The best checkpoint was selected by validation macro-F1, yielding 87.05% accuracy and 0.8245 macro-F1.

3.4 Selective Fine-Tuning

The baseline checkpoint initialised the fine-tuned model. The final residual block (layer4) and classification head (fc) were unfrozen, while earlier convolutional layers remained frozen. This exposed approximately 8.40 million trainable parameters. Differential learning rates allowed the pretrained features to adapt more cautiously than the classification head.

Component	Learning rate
Final residual block (layer4)	0.00001
Classification head (fc)	0.0001

The model was trained for five epochs using AdamW and class-weighted cross-entropy. The best validation macro-F1 (0.9057) was achieved at epoch five, and that checkpoint was selected for the frozen test evaluation.

3.5 Classification Performance Evaluation

Performance was assessed using accuracy, balanced accuracy, per-class precision and recall, F1-score, and confusion matrices. Balanced accuracy averages recall across the seven classes. Macro-F1, which weights classes equally, was the primary model-selection metric because the dataset was imbalanced.

3.6 Predictive Uncertainty Estimation

Normalised predictive entropy was used to quantify the dispersion of the seven-class softmax distribution:

$$H(x) = -[1/\log(7)] \sum_{k=1}^{7} p(x) \log p(x)$$

Entropy is near zero when probability is concentrated on one class and increases as the distribution becomes more uniform. It was treated as an uncertainty-ranking score, not as a complete separation of data ambiguity from model uncertainty.

3.7 Uncertainty-Guided Error Review

Predictions were ranked by descending uncalibrated predictive entropy. Retrospective review budgets of 5%, 10%, 20% and 30% were assessed; 10% was fixed as the primary budget before test evaluation. For each budget, we measured the number of patches flagged, errors captured, error rate among flagged patches, and residual errors among unflagged patches. Error-detection AUROC and AUPRC were also calculated.

This experiment used known labels retrospectively. No pathologist review, prediction correction or prospective clinical workflow was conducted.

3.8 Probability Calibration

Temperature scaling was investigated as a post-hoc method to align predicted confidence more closely with observed correctness. It divides logits by a positive scalar T before softmax:

$$p(y = k \vee x) = \exp(z_i / T) / \sum_{j=1}^{7} \exp(z_j / T)$$

A temperature above one softens the probability distribution without changing the predicted class. T was fitted by minimising negative log-likelihood on approximately half the validation predictions, with the other half used for assessment. The frozen temperature was T = 1.0832799285.

Calibration was assessed using expected calibration error (15 bins), multiclass Brier score and negative log-likelihood. Primary review ranking remained based on uncalibrated entropy, as defined in the frozen protocol.

3.9 High-Confidence Error Analysis

Incorrect validation predictions were examined at maximum softmax confidence thresholds of 90%, 95% and 99%. We recorded the proportion of errors above each threshold and the most frequent true-to-predicted class confusions. This assessed a key limitation of entropy-based triage: some incorrect predictions remain highly confident.

3.10 Frozen Final Test Evaluation

Before accessing held-out test predictions, the evaluation protocol fixed the model checkpoint, checkpoint and split SHA-256 hashes, class ordering, temperature, entropy-ranking method, 10% primary review budget and reported metrics. The final test set contained 13,083 patches. Checkpoint and split integrity were verified before deterministic inference without gradient computation.

No model tuning or calibration fitting was performed using test outcomes. Results were saved as prediction probabilities, per-class metrics, confusion matrix, calibration metrics and review summaries. Because patient and slide identifiers were unavailable, these results represent held-out patch-level performance rather than patient-independent generalisation.

4. Results

4.1 Baseline versus fine-tuned validation performance

The baseline model achieved 87.05% validation accuracy and 0.8245 macro-F1. Selective fine-tuning increased validation accuracy to 92.88% and macro-F1 to 0.9057, corresponding to a macro-F1 gain of 0.0812. Validation errors decreased from 1,694 to 931. This comparison informed model selection and is not an independent test of the benefit of fine-tuning.

Metric	Frozen backbone	Fine-tuned
Validation accuracy	0.8705	0.9288
Validation balanced accuracy	0.8603	0.9183
Validation macro-F1	0.8245	0.9057
Validation errors	1,694	931

4.2 Final held-out patch-level test performance

The selected checkpoint was evaluated once on 13,083 held-out patches after verification of checkpoint and split-file SHA-256 hashes. It correctly classified 12,157 patches and misclassified 926. Test accuracy was 0.9292, balanced accuracy 0.9134 and macro-F1 0.9005. These estimates were close to validation results (accuracy 0.9288; macro-F1 0.9057), suggesting consistency between the two patch-level partitions.

Class	Precision	Recall	F1	Support
FIB	0.9777	0.9460	0.9616	3,943
INF	0.8521	0.9413	0.8945	716
NEC	0.9563	0.9809	0.9685	3,303
NOR	0.8680	0.9066	0.8869	1,980
REA	0.7391	0.8831	0.8047	231
STE	0.9359	0.8796	0.9069	930
TUM	0.9065	0.8566	0.8808	1,980

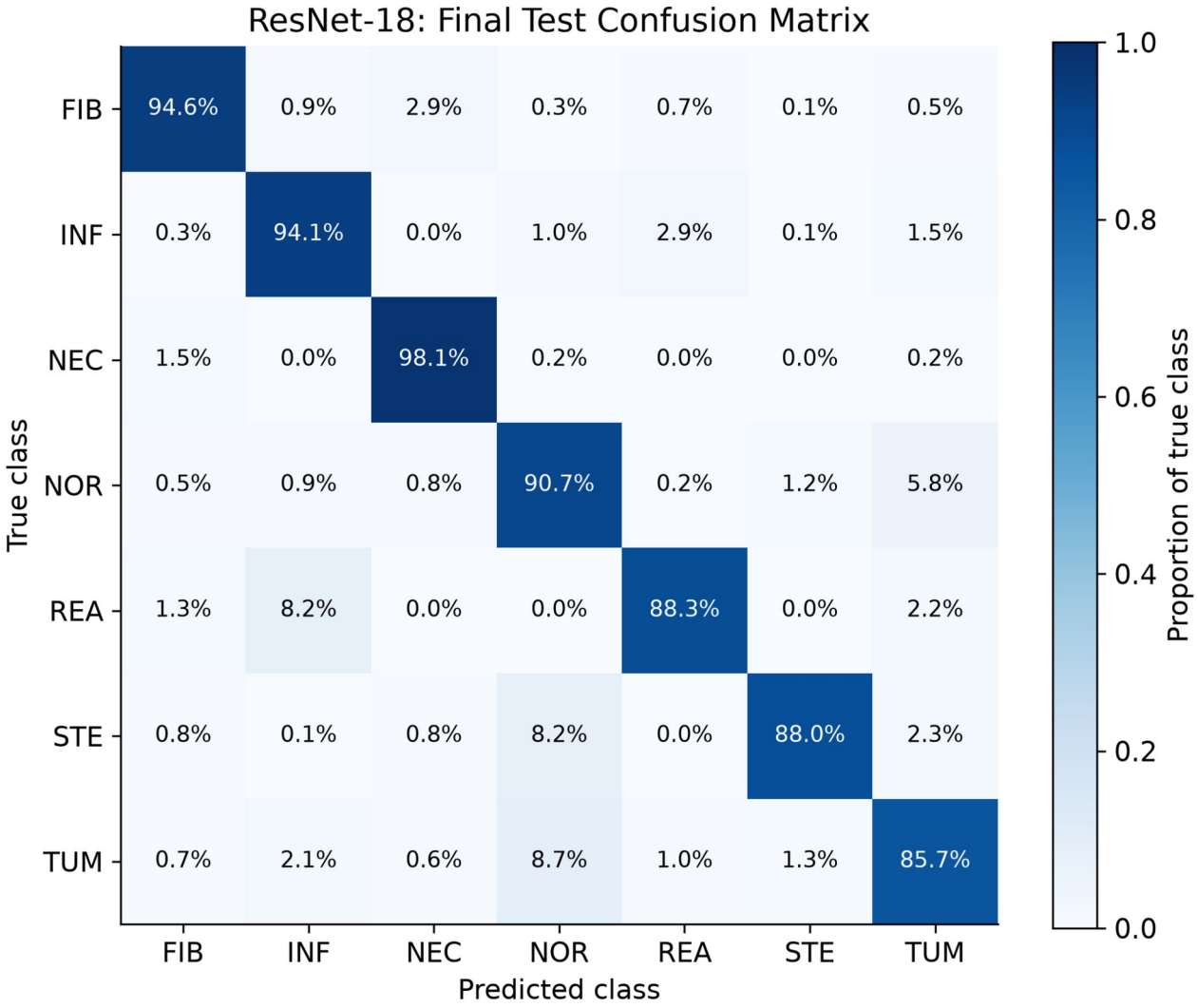


Figure 1. Row-normalised confusion matrix for the frozen final test split. Rows indicate ground-truth categories; columns indicate predicted categories.

Necrosis had the highest recall (98.09%), whereas tumour recall was 85.66%. The most frequent directional confusion was tumour predicted as normal (173 patches); normal predicted as tumour occurred in 114 patches. These two directions accounted for 287 of 926 errors (31.0%). The lowest class F1 was for biliary reaction (0.8047), with only 231 test patches.

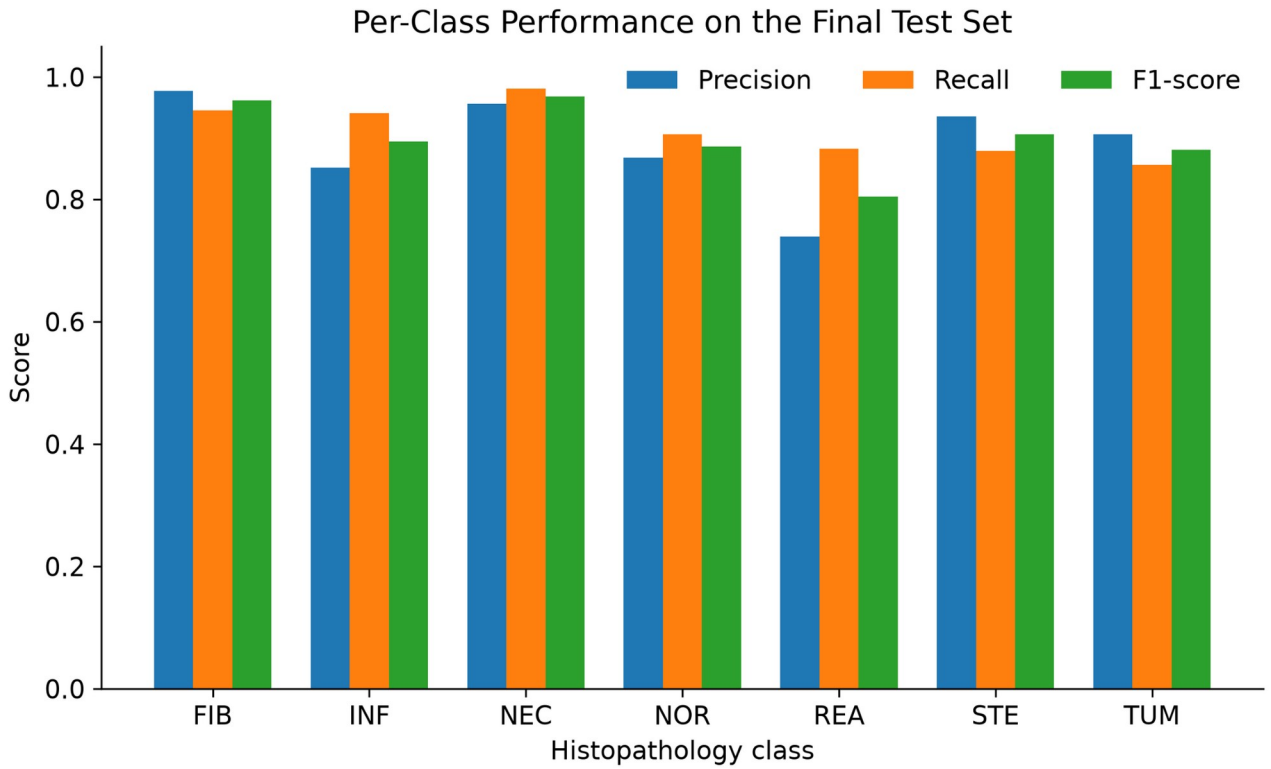


Figure 2. Precision, recall and F1-score by class on the final test split.

4.3 Uncertainty-guided review

Normalised entropy ranked errors effectively (error-detection AUROC 0.9079; AUPRC 0.3928). At the pre-specified 10% review budget, 1,308 patches were flagged, containing 526 of 926 errors (56.80%). The flagged subset had a 40.21% error rate; the 11,775 unflagged patches still contained 400 errors (3.40%). These values measure retrospective prioritisation, not realised improvements from expert correction.

Review budget	Flagged patches	Errors found	Error capture	Unflagged error rate
5%	654	284	30.67%	5.17%
10%	1,308	526	56.80%	3.40%
20%	2,616	755	81.53%	1.63%
30%	3,924	848	91.58%	0.85%

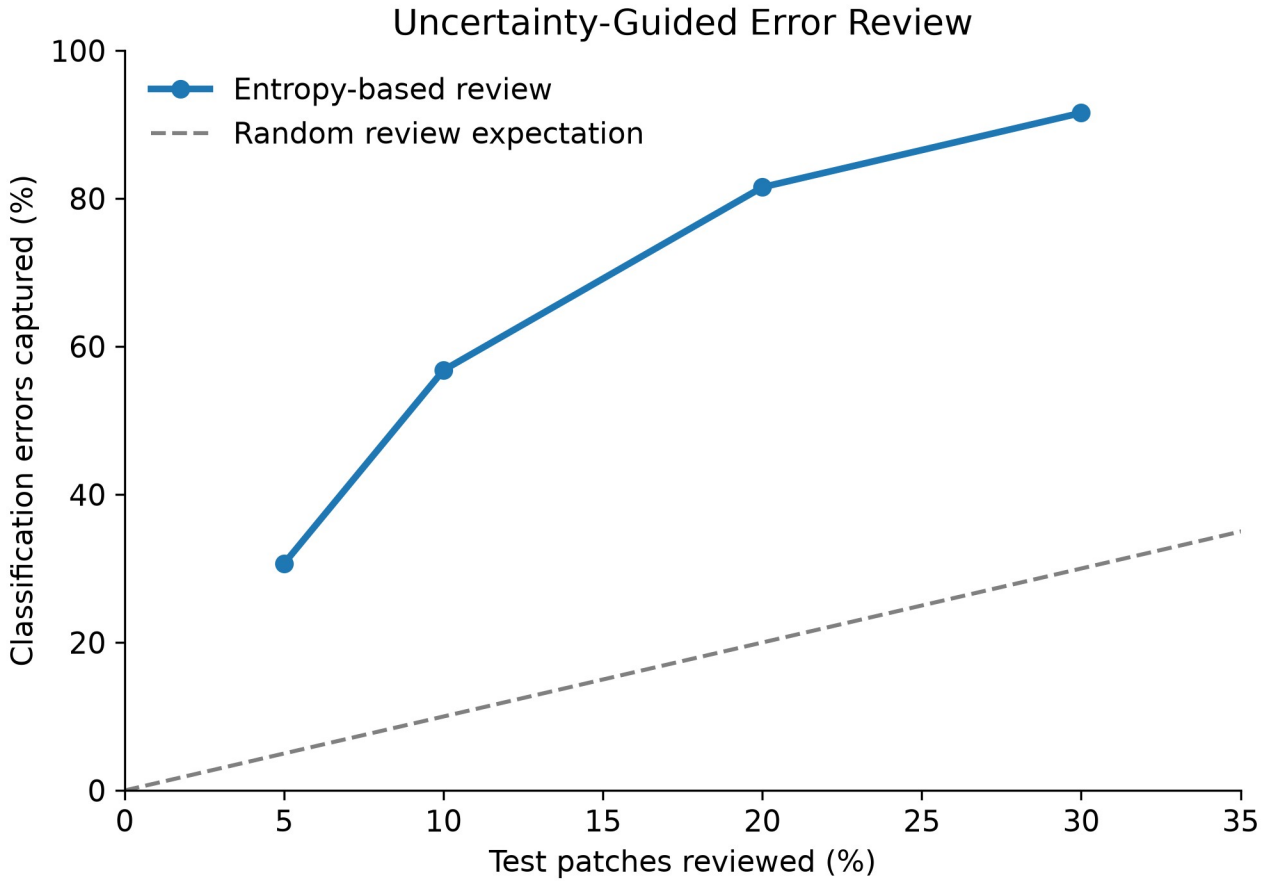


Figure 3. Proportion of classification errors captured by descending-entropy review, compared with the expected proportion under random selection.

4.4 Calibration

Temperature T = 1.0832799285 was fitted on a stratified subset of validation predictions, not on test observations. Applying this fixed temperature reduced test 15-bin ECE from 0.00842 to 0.00617 and negative log-likelihood from 0.20820 to 0.20724. Multiclass Brier score was essentially unchanged (0.10557249 versus 0.10557235). The improvements are small because the uncalibrated model was already well calibrated on average.

Test calibration metric	Uncalibrated	Temperature-scaled
ECE (15 bins)	0.00842	0.00617
Negative log-likelihood	0.20820	0.20724
Multiclass Brier	0.10557249	0.10557235

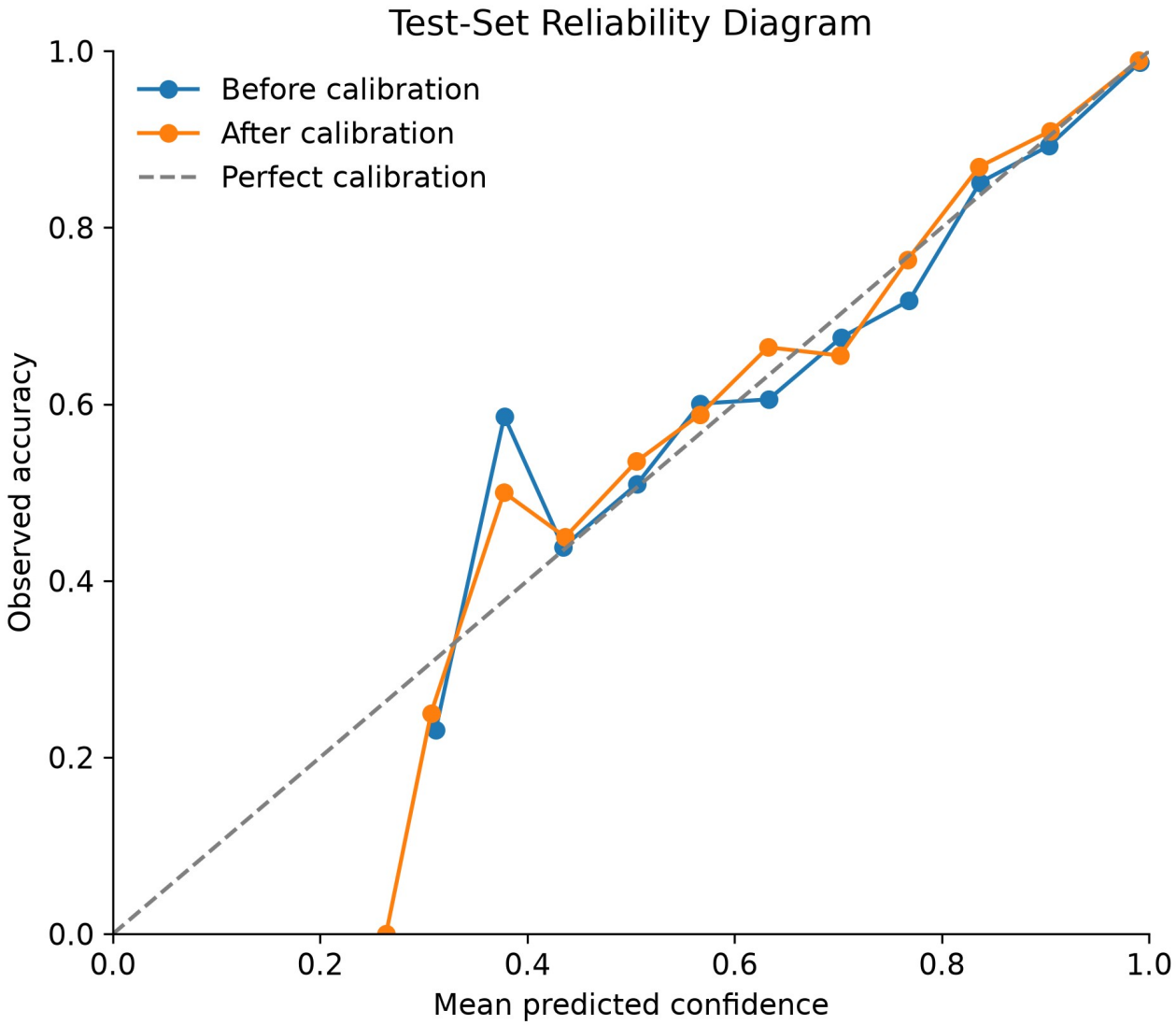


Figure 4. Reliability diagram comparing uncalibrated and temperature-scaled test probabilities. Sparse low-confidence bins can produce noisy empirical accuracies.

4.5 High-confidence validation errors

A retrospective validation analysis found 176 incorrect predictions with maximum softmax confidence at least 0.90, 98 at least 0.95, and 31 at least 0.99. Among the 98 errors above 0.95, 37 had true tumour labels. Thus even a relatively well-calibrated classifier can produce confidently wrong individual predictions. These thresholds were explored on validation data and were not used as the frozen test-review criterion.

5. Discussion

5.1 Interpretation of findings

Selective fine-tuning improved class-balanced validation performance over a frozen-feature baseline, consistent with the value of adapting higher-level features to histopathology. The selected model maintained similar performance on the held-out patch split. Entropy-based ranking concentrated a majority of errors in a small fraction of predictions, offering a potential strategy for prioritising manual inspection in future

experiments. Temperature scaling modestly improved aggregate calibration without changing class predictions.

5.2 Failure modes and class imbalance

Tumour–normal confusion was the dominant error pair, and tumour recall remained below that of fibrosis and necrosis. The biliary reaction class had the lowest F1 and the smallest test support, so its performance estimate is comparatively less stable. The observed patterns are descriptive: without expert review, they cannot be attributed confidently to specific morphological similarities, labelling ambiguity or image artefacts.

5.3 Limitations

- Patient and slide identifiers were unavailable; patch-level splitting may allow source-related patches across partitions and inflate generalisation estimates.
- The validation split was used repeatedly for model selection and exploratory analyses; only the frozen test analysis should be treated as the final held-out evaluation.
- No external cohort, prospective dataset, whole-slide evaluation or clinical reader study was conducted.
- Uncertainty ranking does not catch every error; 400 errors remained outside the 10% highest-entropy test subset.
- The study compared a frozen-backbone baseline with one selectively fine-tuned ResNet-18 configuration; no broad architecture benchmark or statistical uncertainty intervals were established.

5.4 Future research

The most informative next steps would be independently sourced patient- or slide-level evaluation; review of representative errors by qualified pathologists; confidence intervals and repeated-seed experiments on development data; exploration of stain and acquisition shifts; and prospective testing of whether uncertainty prioritisation improves human review efficiency. Such work should use newly designated datasets rather than iteratively tuning against the frozen test split.

6. Conclusion

LiverPath AI provides a reproducible exploratory pipeline for seven-class liver histopathology patch classification, uncertainty ranking and calibration. The fine-tuned ResNet-18 achieved 92.92% accuracy and 0.9005 macro-F1 on 13,083 held-out patches. Reviewing the 10% highest-entropy patches identified 56.80% of test classification errors, while temperature scaling slightly improved ECE. The findings support further investigation of uncertainty-aware computational pathology workflows, but do not demonstrate patient-independent or clinical performance.

7. Reproducibility and data provenance

Dataset: HepatoBench, Hugging Face xtxx/HepatoBench, revision 3fd9d3697fb63a84ea0635aed97f77ad3f1c5d97; reported dataset licence CC BY 4.0 (verify repository terms before redistribution). Image categories are retained exactly as supplied, including STE = steroidal degeneration. Data splits used random seed 42 and SHA-256-based duplicate control. Model: torchvision ResNet-18, ImageNet-pretrained, trained with PyTorch/AdamW and class-weighted cross-entropy. Inference device: Apple Silicon MPS. Final protocol file: results/metrics/final_experiment_protocol.json. Final evaluation: scripts/32_final_test_evaluation.py; figures: scripts/33_generate_final_figures.py. Final output

files: results/final_test/summary.json, predictions.csv, per_class_metrics.csv, confusion_matrix.csv and review_metrics.csv.

Frozen model checkpoint SHA-256: a9e45c3a4f5b2d379b9f5ea28b0a7b9c9f11ce081a0604f2e99d045df2f5b4b1. The original dataset images and checkpoint are not embedded in this report. For public release, document exact dependency versions, data access instructions and any checkpoint redistribution restrictions.

References and resources

He, K., Zhang, X., Ren, S. & Sun, J. (2016). Deep Residual Learning for Image Recognition. Proceedings of CVPR. https://arxiv.org/abs/1512.03385

Guo, C., Pleiss, G., Sun, Y. & Weinberger, K. Q. (2017). On Calibration of Modern Neural Networks. Proceedings of ICML. https://arxiv.org/abs/1706.04599

HepatoBench dataset. Hugging Face repository: https://huggingface.co/datasets/xtxx/HepatoBench (pinned revision specified above).

PyTorch documentation: https://pytorch.org/docs/stable/ ; torchvision models: https://pytorch.org/vision/stable/models.html